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Comparative analysis of nursing care plans produced by artificial intelligence models (ChatGPT, Gemini, and DeepSeek) in terms of readability, reliability, and quality

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Abstract

Background While AI chatbots have increased access to healthcare information, evidence regarding the readability, reliability, and overall quality of nursing care plans generated by these systems remains limited.

Purpose This study aimed to comparatively evaluate nursing care plan texts generated by ChatGPT, Gemini, and DeepSeek in terms of readability, reliability, and overall quality.

Methods Thirty nursing diagnoses were randomly selected from the NANDA International 2021–2023 taxonomy. For each diagnosis, nursing care plans were generated using three AI chatbots, resulting in 90 texts. Outputs were comparatively evaluated using a descriptive information form, the DISCERN instrument, and multiple readability measures (FRES, SMOG, Gunning Fog Index, and Flesch–Kincaid Grade Level).

Results Readability analyses indicated that nursing care plans generated by all three AI models significantly exceeded the recommended sixth-grade reading level ($P < .001$). DISCERN scores reflected moderate reliability, with mean scores of 57.41 ± 5.9 for ChatGPT, 58.41 ± 4.8 for Gemini, and 56.51 ± 6.8 for DeepSeek. Overall, 27 texts (90%) were rated as providing nursing care information of moderate quality. The presence of verifiable references demonstrated a statistically significant positive association with both reliability and quality scores ($P < .05$).

Conclusion Although AI chatbots demonstrate potential as supportive tools in nursing education and documentation, they should not be used as standalone resources for generating complete nursing care plans without professional review. Improvements in content clarity, reference accuracy, and expert oversight are necessary to enhance their applicability in nursing practice.

Keywords Chatbot, ChatGPT, Gemini, DeepSeek, Nursing care, Nursing care plan

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Introduction

The integration of technological innovations in healthcare has the potential to reduce the workload of nurses and physicians while transforming care processes. Among these innovations, artificial intelligence (AI) applications, particularly those based on large language models (LLMs), are increasingly employed for clinical documentation, decision support, and patient education due to their natural language processing capabilities [1]. AI systems can support clinical decision-making, prepare patient education materials, and contribute to the development of nursing care plans, which are critical components of the nursing process [1].

Artificial intelligence encompasses systems that enable computers to perform cognitive functions modeled on human thinking, including decision-making, learning from experiences, analyzing language and visual data, and generating solutions to problems [2]. In nursing practice, AI tools both hardware-based and digital platforms enhance care efficiency and offer novel opportunities for diagnostic and care activities [3–5]. However, to ensure the reliability of AI-generated content, expert review and supervision by experienced healthcare professionals remain essential [1].

The quality of nursing care plans is determined not only by content accuracy but also by readability, clinical applicability, measurability, and alignment with patients' individual needs [6]. Readability refers to how easily a text can be understood and is crucial for healthcare providers as well as patients. Evidence indicates that most patient education materials exceed the recommended sixth-grade reading level, which may hinder comprehension and increase the risk of errors in practice [7, 8]. Reliability is critical, as some AI models can produce hallucinated or fabricated outputs, which may negatively affect clinical decisions and patient safety [9, 10]. Quality also encompasses the clinical relevance of care plans, their measurability, and adherence to standardized frameworks, such as SMART criteria (Specific, Measurable, Achievable, Realistic, Time-bound) [6]. Standardized tools like DISCERN can systematically assess the quality of both written health materials and nursing care plans; in this study, DISCERN was applied to ensure consistent and objective evaluation of AI-generated texts [11].

Several AI models have been developed and implemented in healthcare, among which OpenAI's ChatGPT, Google DeepMind's Gemini, and the China-based DeepSeek have gained significant attention [12]. ChatGPT has been extensively studied due to its widespread adoption in academic and clinical settings [13]. Gemini is notable for its multimodal capabilities, allowing simultaneous processing of textual, visual, and tabular data [14]. DeepSeek, in contrast, is an open-access LLM valued for its high processing speed, cost-effectiveness, and large parameter architecture [15]. Unlike specialized medical models requiring restricted access or proprietary datasets, DeepSeek reflects the type of general-purpose AI tool increasingly encountered by nurses and students in real-world practice. Its inclusion in this study provides a broader comparative perspective and addresses gaps in the literature regarding emerging AI models in nursing care plan generation.

Although previous research has mainly focused on ChatGPT, evaluations of Gemini and DeepSeek remain limited. Furthermore, prior studies have often examined a single dimension, such as readability, reliability, or quality, rather than providing a comprehensive assessment. There is also limited understanding of nurses' perspectives on the clinical validity and applicability of AI-generated nursing care plans [16]. This study addresses these gaps by comparatively evaluating nursing care plans generated by ChatGPT, Gemini, and DeepSeek in terms of readability, reliability, and quality.

Ensuring high readability, reliability, and quality of AI-generated nursing care plans is essential for supporting nurses' workflow, promoting evidence-based practice, and safeguarding patient safety. The findings of this research aim to provide practical insights into the advantages and limitations of various AI systems, supporting informed integration of AI into clinical nursing practice and education.

Materials and methods

Research design and sample

This study was designed as an exploratory and descriptive comparative analysis aimed at evaluating the readability, reliability, and quality of nursing care plan texts generated by different artificial intelligence (AI) chatbots. Consistent with established frameworks for the evaluation of large language models (LLMs) in healthcare [16], the primary objective was to conduct a structured comparison of AI-generated outputs under standardized conditions rather than to achieve statistical generalizability or population-level inference.

The study sample consisted of 30 nursing diagnoses randomly selected from the NANDA International taxonomy (2021–2023). For each diagnosis, nursing care plan texts were generated using three AI models ChatGPT (OpenAI), Gemini (Google), and DeepSeek resulting in a total of 90 AI-generated texts ($N = 90$) [17].

The selection of 30 nursing diagnoses as the sampling unit aligns with prior exploratory studies assessing AI-generated health-related content, in which similar sample sizes were considered sufficient to identify meaningful patterns, linguistic characteristics, and comparative differences between AI models [18, 19]. The sample size was sufficient to explore differences across AI models but was not intended to support population-level generalizations.

Given the exploratory and descriptive nature of the study, the sample size was determined based on feasibility and comparative adequacy rather than statistical power calculations.

Accordingly, the application of the Central Limit Theorem was not required for this study, as no assumptions of sampling distributions or population-level parameter estimation were made. The selected sample size was deemed sufficient to support model-level comparisons and descriptive analyses of AI-generated nursing care plan texts under controlled and standardized conditions.

AI models and text generation procedure

Three large language model–based artificial intelligence chatbots were evaluated in this study: ChatGPT (OpenAI), Gemini (Google), and DeepSeek. These models were deliberately selected because they represent widely used AI systems with distinct provider ecosystems, access conditions, underlying architectures, and functional capabilities, including differences in parameter exposure and multimodal potential. All text generations were conducted on July 8, 2025, using publicly accessible web-based interfaces [20–22].

ChatGPT was accessed via the web interface using the freely available version provided by OpenAI at the time of data collection [20]. Gemini was accessed through Google's web-based Gemini platform [21], and DeepSeek was accessed via its publicly available web interface [22]. All three tools were used through free web-based access.

All outputs were generated in English. A standardized prompt was used for all models to ensure comparability. Each nursing diagnosis was processed in a separate, new session to minimize contextual carryover. For each diagnosis, a single nursing care plan output was generated per model. Only the first complete output generated in each session was included in the analysis; no further selection, filtering, or reranking was performed in order to avoid selection bias.

All models were used with their default system settings. Model-specific parameters such as temperature or maximum token limits were not manually adjusted, as these options were not explicitly exposed to the user through the web interfaces at the time of data collection [20–22].

Because the models were accessed exclusively through publicly available web-based interfaces, detailed internal model identifiers and system-level parameters were not disclosed by the providers; therefore, reproducibility was ensured through standardized prompts, fixed access conditions, explicit reporting of generation rules, and precise documentation of the data collection date.

Selection and randomization of nursing diagnoses

The texts produced by AI tools (ChatGPT, Gemini, and DeepSeek) were derived from topics chosen from the 2021–2023 NANDA International taxonomy, covering a total of 267 unique nursing diagnoses [18].

With all 267 nursing diagnoses listed, a randomization table was produced using https://www.random.org (refer to Supplementary Digital Content Randomization Table.docx). This method allowed 30 diagnoses to be chosen at random, ensuring equal selection probability for each and preserving the fairness of the sampling procedure [23, 24]. Each selected diagnosis was then input into three AI chatbots to create care plans, producing a total of 90 texts.

Data collection

A structured data collection protocol was implemented to comparatively evaluate the performance of current AI models in producing texts consistent with academic conventions. Data collection was conducted on July 8, 2025, using the web-based interfaces of the latest publicly available versions of ChatGPT [20], Gemini [21], and DeepSeek [22]. All platforms were accessed via Google Chrome on a Windows operating system to ensure reproducibility.

For each randomly selected nursing diagnosis, the AI chatbots (ChatGPT, Gemini, and DeepSeek) were provided with the prompt: "Prepare a text on -NANDA DIAGNOSIS- and the corresponding nursing care plan, including as many references and in-text citations as you can." The aim of this prompt was twofold: to generate meaningful content and to evaluate the models' capacity to include academic-style references [8]. Emphasizing the term "nursing care plan" was intended to ensure that the generated texts were clinically relevant and logically structured [25, 26]. Assessing AI models' ability to produce texts containing academic citations represents an emerging area of research interest [27].

All texts generated by ChatGPT, Gemini, and DeepSeek were systematically saved by the investigators for subsequent evaluation and analysis.

Data collection tools

Descriptive information assessment

The descriptive data collection form comprised 12 items covering characteristics such as word and paragraph count, text similarity score (measured with iThenticate and Turnitin), number of references, year of publication, type of document, and access options [28, 29].

Readability assessment

Two online platforms were used to examine the readability of nursing care plan texts generated by AI chatbots (ChatGPT, Gemini and DeepSeek). ReadabilityFormulas.com (Calculator 1) and Online-Utility.org (Calculator 2) employ various readability formulas to systematically evaluate the texts [30–32]. These seven formulas were

selected because they are among the most widely cited and validated readability measures in healthcare and patient education research.

Seven established readability formulas were applied in the analysis: Flesch Reading Ease Score (FRES), Flesch–Kincaid Grade Level (FKGL), Gunning Fog Index (GFOG), Simple Measure of Gobbledygook (SMOG), Automated Readability Index (ARI), Coleman–Liau Index (CLI), and Linsear Write (LW). Each text's readability was calculated using these formulas, and results were reported as medians with minimum and maximum values.

Prior to readability analysis, all AI-generated nursing care plan texts were pre-processed to ensure consistency across calculations. Section headings, subheadings, bullet points, numbering, and reference lists were removed, and only the main narrative content of each care plan was entered into the readability calculators. This approach was used to prevent distortion of readability scores by non-narrative text elements.

Readability thresholds were defined in line with widely accepted health communication guidelines, which recommend that health-related materials be written at or below a sixth-grade reading level to ensure accessibility and comprehension [33, 34]. For grade-level–based formulas (FKGL, GFOG, SMOG, ARI, CLI, and LW), lower scores indicate easier readability. Accordingly, a score of ≤ 6 was operationalized as meeting the sixth-grade readability standard, whereas scores > 6 were interpreted as exceeding the recommended readability level.

In contrast, the Flesch Reading Ease Score (FRES) operates on an inverse scale, with higher scores reflecting easier readability. Consistent with established interpretations, FRES values ≥ 80 were considered indicative of sixth-grade readability, while lower scores indicated increased text difficulty [30, 33].

In the statistical analyses, the median readability scores obtained for each AI model were compared with these predefined sixth-grade thresholds. Texts with median values > 6 for grade-level formulas or < 80 for FRES were classified as not meeting the recommended sixth-grade readability standard. The calculation methods of the seven applied readability formulas and related information are presented in Supplementary Materials.

This clarification corrects earlier ambiguities in the directionality of grade-level thresholds and ensures consistency across all applied readability formulas.

Reliability and quality assessment (DISCERN)

The reliability and quality of the AI-generated nursing care plan texts were evaluated using the DISCERN instrument by two independent nursing researchers with experience in nursing education and research. Prior to data collection, both evaluators reviewed the DISCERN handbook and scoring guidelines and jointly assessed a small set of pilot texts to ensure a shared understanding of the evaluation criteria, as recommended in methodological guidelines for DISCERN-based assessments [11].

Each evaluator independently assessed all 90 texts using the standardized DISCERN scoring procedure. All AI-generated texts were anonymized prior to evaluation, and any identifiers indicating the source AI model were removed. The evaluators were blinded to the originating model (ChatGPT, Gemini, or DeepSeek) to minimize potential observer bias. Evaluations were conducted independently and without access to the other evaluator's scores.

In cases of scoring discrepancies between the two evaluators, the arithmetic mean of the two scores was calculated and used as the final DISCERN score for statistical analysis. This approach has been widely adopted in previous studies evaluating the reliability and quality of online and AI-generated health information using the DISCERN instrument [8].

To enhance transparency in inter-rater reliability assessment, different agreement statistics were applied according to data type. Cohen's kappa (κ) was calculated for categorical quality classifications derived from DISCERN total scores (low, moderate, high). Intraclass correlation coefficients (ICC) were computed for continuous DISCERN total scores using a two-way random-effects model with absolute agreement (ICC [2,1]), as both raters were considered representative of a larger population of evaluators. No third-rater adjudication was performed, as a consensus approach was not required for this exploratory analysis.

Statistical analysis

All statistical analyses were performed using SPSS version 25 (IBM Corp., Armonk, NY). Descriptive statistics were summarized as medians and interquartile ranges for continuous variables and as counts and percentages for categorical variables.

Given that each nursing diagnosis was evaluated across all three AI models, the data exhibited a repeated-measures structure. Accordingly, non-parametric Friedman tests were used to compare readability, reliability, and quality scores across ChatGPT, Gemini, and DeepSeek. When the Friedman test indicated statistically significant differences, post-hoc pairwise comparisons were conducted using Wilcoxon signed-rank tests with Bonferroni correction to control for multiple comparisons. Because observations were not independent across models for the same nursing diagnosis, the Kruskal–Wallis H test was not appropriate; therefore, the Friedman test was selected as the correct non-parametric alternative for repeated-measures comparisons. Non-parametric methods were

chosen due to the repeated-measures design and the non-normal distribution of the data.

Associations between numerical variables and DISCERN scores were examined using Spearman's rank correlation coefficient. The reliability of readability assessment tools was evaluated using intraclass correlation coefficients (ICC), and inter-rater agreement was assessed using Cohen's kappa (κ). Text similarity was assessed using Turnitin and iThenticate software. A p-value < 0.05 was considered statistically significant. All statistical analyses were conducted for descriptive and comparative purposes, and the findings should be interpreted within the exploratory scope of the study. Inter-rater agreement for categorical quality classifications derived from DISCERN total scores (low, moderate, high) was assessed using Cohen's kappa (κ), which is recommended for evaluating agreement between independent raters for categorical data [35].

The agreement between evaluators for continuous DISCERN total scores was examined using intraclass correlation coefficients (ICC). A two-way random-effects model with absolute agreement (ICC [2,1]) was applied, as both raters were considered representative of a larger population of potential evaluators and the absolute concordance between ratings was of interest [36, 37].

These reliability statistics ensured the robust integration of DISCERN scores into subsequent comparative analyses across AI models.

ICC values were interpreted according to established guidelines, with values below 0.50 indicating poor agreement, values between 0.50 and 0.75 indicating moderate agreement, 0.75–0.90 indicating good agreement, and values above 0.90 indicating excellent agreement [37].

Text similarity was assessed using Turnitin and iThenticate software. A p-value < 0.05 was considered statistically significant. All analyses were conducted for descriptive and comparative purposes within the exploratory scope of the study.

Ethical considerations

This study did not involve human participants, patient data, or any identifiable personal information. All analyses were conducted exclusively on AI-generated nursing care plan texts created for research purposes. As no human subjects or identifiable data were involved, institutional ethical committee approval and informed consent were not required, in accordance with institutional and journal guidelines.

No real patient records, clinical data, or confidential information were used at any stage of the study. The AI platforms were accessed through publicly available web-based interfaces. All AI-generated data were fully anonymized, securely stored, and transparently reported to ensure methodological rigor, data transparency, and ethical integrity.

References generated within the AI-produced nursing care plan texts were treated strictly as part of the study data and were not considered authoritative sources. These AI-generated citations were not included in the reference list of this manuscript. The reference list reflects only sources independently selected, verified, and cited by the authors.

Results

Descriptive characteristics of AI-generated texts

Two researchers independently evaluated nursing care plan texts generated by ChatGPT, Gemini, and DeepSeek based on randomly selected nursing diagnoses. Overall, the length of the generated texts was comparable across models, with mean word counts of approximately 500 words. Gemini produced slightly longer texts, whereas DeepSeek outputs tended to be shorter and more variable in length (Table 1. Descriptive Characteristics of AI-Generated Nursing Care Plan Texts).

Source analysis revealed a critical finding: the majority of references cited within the AI-generated care plans were either fictional or could not be retrieved from academic databases. Although traditional plagiarism detection tools (iThenticate and Turnitin) reported a 0% similarity rate for all texts classifying them as fully original Turnitin's AI detection module identified all texts as AI-generated ($n = 90$; 100%). This discrepancy highlights an important limitation of conventional plagiarism tools, which are unable to detect fabricated yet original-looking references, thereby posing a potential risk to academic validity and clinical reliability.

Readability analyses demonstrated that nursing care plans generated by all three AI models significantly exceeded the recommended sixth-grade reading level across all readability indices (Friedman test, $p < .001$). This indicates that the texts require advanced reading skills and may be difficult to understand for users with limited health literacy.

Although detailed readability scores are presented in Tables 2, 3 and 4, the overall pattern was consistent: DeepSeek-generated texts showed the highest linguistic complexity, followed by ChatGPT, while Gemini produced relatively more readable outputs. However, even Gemini-generated texts remained above the recommended readability threshold (Table 2. Readability scores of nursing care plan texts generated by ChatGPT, Gemini, and DeepSeek, and analysis of the text content at a 6th-grade reading level [median (minimum–maximum)], were performed using Calculator 1.,Table 3. Readability scores of nursing care plan texts generated by ChatGPT, Gemini, and DeepSeek, and analysis of the text content at a 6th-grade reading level [median

Table 1 Descriptive characteristics of AI-generated nursing care plan texts

	ChatGPT		Gemini		DeepSeek	
Variables	Min-Max	Mean ± SD	Min-Max	Mean ± SD	Min-Max	Mean ± SD
Publication Years of References	1984–2023	2018 ± 5.05	1979–2023	2016 ± 7.00	1980–2023	2017 ± 6.05
Number of References	4–7	5.6 ± 1.08	3–8	4.33 ± 2.06	3–9	4.36 ± 1.91
References that Are Accessible	0–4	0.63 ± 1.01	0–5	0.53 ± 1.01	0–3	0.63 ± 1.01
References from the Web	1–4	1.14 ± 1.88	1–4	2.14 ± 1.58	1–7	2.14 ± 1.88
References from the Web That Are Accessible	1–3	0.25 ± 0.58	1–3	0.27 ± 0.58	1–4	0.25 ± 1.58
References Cited from Journals	1–6	2.25 ± 1.25	1–6	2.35 ± 1.25	1–3	2.25 ± 0.25
Number of Available Journal Article References	1–3	0.23 ± 0.58	1–3	0.23 ± 0.60	1–3	0.23 ± 0.58
References from Books	1–2	1.46 ± 0.36	1–2	1.56 ± 0.32	1–2	1.66 ± 0.29
Book References That Are Accessible	1–2	0.25 ± 0.59	1–2	0.25 ± 0.59	1–2	0.25 ± 0.59
Word Count	402–741	495.3 ± 134.4	487–587	531 ± 41.1	374–552	497.7 ± 70.1
Number of Paragraphs	4–9	5.65 ± 1.00	4–8	5.65 ± 1.00	4–6	5.65 ± 1.00
iThenticate Similarity Rate (%)	-	0	-	0	-	0
Turnitin Similarity Rate (%)	-	0	-	0	-	0
Turnitin AI Similarity Rate (AI-generated, n [%])	-	30 (100%)	-	30 (100%)	-	30 (100%)

Note: All texts showed 0% similarity in plagiarism detection tools (iThenticate and Turnitin). Turnitin AI detection classified all generated texts as AI-generated (100%), resulting in a ceiling effect; therefore, results are presented as frequencies and percentages rather than mean ± standard deviation

(minimum–maximum)], were performed using Calculator 2.,Table 4. The analysis of readability metrics of nursing care plan texts generated by ChatGPT, Gemini, and DeepSeek, and the statistical evaluation of text complexity according to a 6th-grade reading standard [median (minimum–maximum)], was performed using mean values obtained from Calculator 1 and Calculator 2 for comparison purposes.).

From a practical perspective, exceeding the sixth-grade reading level suggests limited usability for both nursing students and patients. Complex sentence structures and advanced terminology may hinder rapid comprehension in clinical settings, where clarity and efficiency are essential for effective nursing care planning.

Reliability and quality of AI-generated texts based on DISCERN scores

Assessment using the DISCERN instrument revealed that AI-generated nursing care plans demonstrated moderate reliability and moderate informational quality across all models (Table 5. Mean Scores of the DISCERN Total Scale and Subscales and Evaluation of Texts Generated by ChatGPT, Gemini, and DeepSeek Using DISCERN). While all texts were classified within the moderate reliability category, this finding should be interpreted cautiously, as it reflects consistency rather than high clinical trustworthiness.

Only a small proportion of texts achieved high overall quality scores (10% for ChatGPT; 6.6% for Gemini and DeepSeek), whereas approximately one-fifth were rated as low quality. This distribution indicates that, although AI-generated care plans may provide a general framework, they frequently lack the depth, specificity, and clinical precision required for direct implementation in nursing practice.

Importantly, texts that included a higher number of accessible and verifiable references demonstrated significantly greater reliability and higher DISCERN scores. This finding suggests that reference transparency plays a critical role in enhancing the perceived trustworthiness of AI-generated nursing care plans. In practical terms, unverifiable references may mislead users and undermine confidence in AI-supported clinical documentation.

Reliability of readability calculators (Intraclass correlation coefficients)

Readability scores calculated using two independent online tools showed excellent inter-calculator reliability for all indices and AI models (Table 6. Correlation Analysis Between Descriptive Criteria and DISCERN). This finding confirms that the observed readability results are robust and not dependent on a specific measurement platform.

Inter-rater reliability of DISCERN assessments

Inter-rater reliability analyses demonstrated strong agreement between the two evaluators. Cohen's kappa coefficient for categorical DISCERN classifications indicated excellent agreement ($\kappa = 0.827$). Similarly, intraclass correlation coefficients for continuous DISCERN total scores showed good to acceptable agreement across all AI models, supporting the consistency and reproducibility of the quality assessments.

Discussion

In this study, nursing care plan texts generated by three widely used AI chatbots (ChatGPT, Gemini, and DeepSeek) were evaluated in terms of readability, quality, and reliability. The findings indicate that although AI-generated content is increasingly used in healthcare-related

Table 2 Readability scores of nursing care plan texts generated by ChatGPT, Gemini, and DeepSeek, and analysis of the text content at a 6th-grade reading level [median (minimum–maximum)], were performed using calculator 1. (https://readabilityformulas.com/free-readability-formula-tests.php)

Calculator 1 statistics	ChatGPT	Google Gemini	DeepSeek	ChatGPT C6th-GRL (*P*)*1	Gemini C6th-GRL (*P*)*1	DeepSeek C6thGRL (*P*)*1	Between Chat-GPT and Gemini2	Between Chat GPT and DeepSeek2	Between Gemini and Deep-Seek2
FRES	31.35 (9–55)	40.50 (22–55)	36 (10–70)	**<0.001**	**<0.001**	**<0.001**	0.005	0.830	0.026
GFOG	17.75(13.40–25.65)	15.78 (12–25.09)	20.47 (15.60–54.60)	**<0.001**	**<0.001**	**<0.001**	0.007	0.001	**<0.001**
FKGL	14.29 (11.74–20.72)	11.23 (7.79–15.99)	15.99 (13.13–49.57)	**<0.001**	**<0.001**	**<0.001**	0.001	0.001	**<0.001**
CLI	15.17 (12.37–21.42)	13.71 (11.08–16.28)	15.87 (12.31–18.63)	**<0.001**	**<0.001**	**<0.001**	0.008	0.213	**<0.001**
SMOG	12.85(9.57–18.09)	11.67 (8.67–16.42)	14.58 (11.57–32.37)	**<0.001**	**<0.001**	**<0.001**	0.005	0.002	**<0.001**
ARI	16.52 (12.58–23.45)	14.47 (10.14–26.16)	18.71 (14.15–60.62)	**<0.001**	**<0.001**	**<0.001**	0.015	0.001	**<0.001**
LW	16.27 (12.44–22.38)	14.32 (7.42–31.10)	23.41 (14.50–85.25)	**<0.001**	**<0.001**	**<0.001**	0.017	**<0.001**	**<0.001**
Grade level	15.00 (12.00–20.00)	13.00 (10.00–20.00)	18.00 (13.00–45.00)	**<0.001**	**<0.001**	**<0.001**	0.022	0.005	**<0.001**
Reading level	n (%)	n (%)	n (%)						
Difficult to read	1 (4.6)	5 (22.7)	0 (0)				**<0.001**	**<0.001**	**<0.001**
Very difficult to read	3 (12.6)	6 (27.3)	1 (4.5)						
Extremally difficult to read	13 (58.1)	7 (31.8)	19 (86.4)						
Professional	5 (23.7)	3 (13.6)	2 (9.1)						
Somewhat difficult	0 (0)	1 (4.5)	0 (0)						
Readers age	n (%)	n (%)	n (%)						
8–9 years old (Fourth and Fifth graders)	0 (0)	1 (4.5)	0 (0)				0.019	0.255	<0.001
10–11 years old (Fifth and Sixth graders)	0 (0)	0 (0)	0 (0)						
11–13 years old (Sixth and Seventh graders)	0 (0)	0 (0)	0 (0)						
12–14 years old (Seventh and Eighth graders)	0 (0)	0 (0)	0 (0)						
13–15 years old (Eighth and Ninth graders)	0 (0)	0 (0)	0 (0)						
14–15 years old (Ninth to Tenth graders)	0 (0)	0 (0)	0 (0)						
15–17 years old (Tenth to Eleventh graders) n (%)	0 (0)	1 (4.5)	0 (0)						
17–18 years old (Twelfth graders)	2 (4.6)	4 (19.2)	0 (0)						

Table 2 (continued)

Calculator 1 statistics	ChatGPT	Google Gemini	DeepSeek	ChatGPT C6th-GRL (*P*)*1	Gemini C6th-GRL (*P*)*1	DeepSeek C6thGRL (*P*)*1	Between Chat-GPT and Gemini2	Between Chat GPT and DeepSeek2	Between Gemini and Deep-Seek2
18–19 years old (college level entry)	2 (13.7)	6 (27.4)	2 (4.4)						
21–22 years old (college level)	4 (21.7)	4 (13.6)	3 (9.2)						
23 + years old	14 (59.2)	8 (31.8)	16 (72.8)						
College graduate	0 (0)	0 (0)	3 (12.7)						

P values in bold are statistically significant

ARI = Automated Readability Index, CLI = Coleman-Liau Index, FKGL = Flesch-Kincaid Grade Level, FRES = Flesch reading ease score, GFOG = Gunning FOG, LW = Linsear Write, SMOG = Simple Measure of Gobbledygook

*C6thGRL (P): Comparison of the responses according to 6th grade reading level (P)

[1]Friedman test for repeated-measures comparisons among ChatGPT, Gemini, and DeepSeekWhen statistically significant differences were detected, post-hoc pairwise comparisons were performed using Wilcoxon signed-rank tests with Bonferroni correction

[2]Chi-square test was used for categorical variables

contexts, significant limitations remain when these tools are applied to nursing care planning.

The U.S. National Institutes of Health recommends that health-related materials be written at or below a sixth-grade reading level to ensure accessibility. However, the nursing care plans generated by all three AI chatbots consistently exceeded this threshold. This finding aligns with previous research reporting that AI-generated health information often surpasses recommended readability levels across various clinical domains. Ozduran et al. (2025) demonstrated that AI-generated responses to pain-related questions were difficult to understand and of limited quality, while Nutbeam (2023) emphasized that outputs from AI platforms frequently exceed the comprehension capacities of individuals with limited health literacy [38, 39]. Similar results have been reported for subdural hematoma and cancer-related questions, where AI-generated responses were found to be linguistically complex and challenging for users with lower health literacy [40, 41].

The practical implications of these findings are substantial. Complex language and dense technical terminology may hinder nurses' and nursing students' ability to interpret and implement nursing care plans effectively. Moreover, excessive linguistic complexity may limit patient understanding and reduce active participation in care processes, potentially affecting adherence to treatment plans. Therefore, ensuring that AI-generated nursing care plans are clear, concise, and aligned with recommended literacy levels is essential for supporting both clinical practice and patient-centered care.

Beyond readability, the quality of nursing care plans depends on structured organization, clinical relevance, alignment with standardized nursing frameworks, and the inclusion of measurable and achievable goals. In the present study, nursing care plans generated by ChatGPT, Gemini, and DeepSeek were generally evaluated as being of moderate quality. Although many texts contained clinically relevant information, they frequently lacked specificity, measurable outcomes, and consistent alignment with established frameworks. These findings are consistent with previous studies reporting that AI-generated nursing care plans often remain generic and insufficiently structured when evaluated using standardized nursing classifications [4, 42]. This observation underscores that information accuracy alone does not guarantee high-quality nursing documentation; content must also be practical, understandable, and actionable in real clinical settings.

It is important to note that some limitations in text quality may be attributable to the prompt design used in this study. A zero-shot and intentionally generic prompt was employed without role assignment or explicit structural guidance. While this approach ensured

Table 3 Readability scores of nursing care plan texts generated by ChatGPT, Gemini, and DeepSeek, and analysis of the text content at a 6th-grade reading level [median (minimum–maximum)], were performed using calculator 2. (https://readabilityformulas.com/free-readability-formula-tests.php)

Calculator 2 statistics	ChatGPT	Google Gemini	DeepSeek	ChatGPT C6thGRL (*P*)*1	Gemini C6th-GRL (*P*)*1	DeepSeek C6thGRL (*P*)*1	Between Chat-GPT and Gemini2	Between ChatGPT and DeepSeek2	Between gemini and Deep-Seek2
FRES	28.73 (9.59–61.29)	40.83 (24.59–53.29)	33.83 (9.59–72.29)	**< 0.001**	**< 0.001**	**< 0.001**	0.003	0.649	0.033
GFOG	16.53 (13.12–26.16)	13.43 (10.12–21.11)	19.04 (14.12–51.16)	**< 0.001**	**< 0.001**	**< 0.001**	0.004	0.003	**< 0.001**
FKGL	14.98 (11.25–21.04)	11.98 (7.25–16.04)	16.98 (13.25–48.04)	**< 0.001**	**< 0.001**	**< 0.001**	0.001	0.004	**< 0.001**
CLI	14.83 (11.43–21.30)	13.98 (11.25–16.04)	15.98 (12.25–18.04)	**< 0.001**	**< 0.001**	**< 0.001**	0.003	0.140	**< 0.001**
SMOG	16.31 (12.70–20.46)	13.38 (11.25–16.24)	16.98 (14.25–32.04)	**< 0.001**	**< 0.001**	**< 0.001**	0.005	0.003	**< 0.001**
ARI	16.14 (11.71–21.74)	13.28 (10.25–20.24)	17.88 (13.25–59.04)	**< 0.001**	**< 0.001**	**< 0.001**	0.006	0.001	**< 0.001**

P values in bold are statistically significant

ARI = Automated Readability Index, CLI = Coleman-Liau Index, FKGL = Flesch-Kincaid Grade Level, FRES = Flesch reading ease score, GFOG = Gunning FOG, LW = Linsear Write, SMOG = Simple Measure of Gobbledygook

*C6thGRL (P): Comparison of the responses according to 6th grade reading level (P)

[1] Friedman test for repeated-measures comparisons among ChatGPT, Gemini, and DeepSeekWhen statistically significant differences were detected, post-hoc pairwise comparisons were performed using Wilcoxon signed-rank tests with Bonferroni correction

[2] Chi-square test was used for categorical variables

Table 4 The analysis of readability metrics of nursing care plan texts generated by ChatGPT, Gemini, and DeepSeek, and the statistical evaluation of text complexity according to a 6th-grade reading standard [median (minimum–maximum)], was performed using mean values obtained from calculator 1 and calculator 2 for comparison purposes

Calculator 1 statistics	ChatGPT *	ChatGPT C6thGRL (*P*)[1,2]	Gemini*	Gemini C6thGRL (*P*)[1,2]	DeepSeek*	DeepSeek C6thGRL (*P*)[1,2]	Between ChatGPT and Gemini	Between Chat-GPT and DeepSeek	Between gemini and DeepSeek
FRES	30.04 (9.29–58.14)	**< 0.001**	40.66 (23.29–54.14)	**< 0.001**	34.91 (9.79–71.14)	**< 0.001**	0.003	0.745	0.021
GFOG	17.14 (13.26–25.9)	**< 0.001**	14.60 (11.06–23.1)	**< 0.001**	19.75 (14.86–52.88)	**< 0.001**	0.004	0.003	**< 0.001**
FKGL	14.63 (11.49–20.88)	**< 0.001**	11.60 (7.52–16.01)	**< 0.001**	16.48 (13.19–48.80)	**< 0.001**	0.001	0.003	**< 0.001**
CLI	15.00(11.09–21.36)	**< 0.001**	13.84(11.16–16.16)	**< 0.001**	15.92(12.28–18.33)	**< 0.001**	0.003	0.001	**< 0.001**
SMOG	14.58 (11.13–19.27)	**< 0.001**	12.52 (9.96–16.33)	**< 0.001**	15.78 (12.91–32.2)	**< 0.001**	0.015	0.105	**< 0.001**
ARI	16.33 (12.14–22.59)	**< 0.001**	13.87 (10.19–23.2)	**< 0.001**	18.29 (13.7-59.83)	**< 0.001**	0.006	0.002	**< 0.001**

Calculator 1: https://readabilityformulas.com/free-readability-formula-tests.php

Calculator 2: https://www.online-utility.org/english/readability_test_and_improve.jsp

P values in bold are statistically significant

ARI = Automated Readability Index, CLI = Coleman-Liau Index, FKGL = Flesch-Kincaid Grade Level, FRES = Flesch reading ease score, GFOG = Gunning FOG, LW = Linsear Write, SMOG = Simple Measure of Gobbledygook

*[(Calculator 1) + (Calculator 2)]/21C6thGRL (P): Comparison of the responses according to 6th grade reading level (P)

2Friedman test for repeated-measures comparisons among ChatGPT, Gemini, and DeepSeekWhen statistically significant differences were detected, post-hoc pairwise comparisons were performed using Wilcoxon signed-rank tests with Bonferroni correction3Chi-square test was used for categorical variables

Table 5 Mean scores of the DISCERN total scale and subscales and evaluation of texts generated by ChatGPT, Gemini, and deepseek using DISCERN

DISCERN Sections	ChatGPT Min-Max	ChatGPT Mean ± SD	Chat-GPT Poor (Score < 40%)	ChatGPT Fair (Score 40%-79%)	Chat-GPT Good (Score > 79%)	Gemini Min-Max	Gemini Mean ± SD	Gemini Poor (Score < 40%)	Gemini Fair (Score 40%-79%)	Gemini Good (Score > 79%)	DeepSeek Min-Max	DeepSeek Gemini Mean ± SD	DeepSeek Poor (Score < 40%)	DeepSeek Fair (Score 40%-79%)	DeepSeek Good (Score > 79%)
S1 to S8 Reliability of Publication*	21–35	26.62 ± 2.85	0 (0)	30 (100)	0 (0)	20–35	25.62 ± 3.85	0 (0)	20 (100)	0 (0)	20–35	25.62 ± 3.85	0 (0)	20 (100)	0 (0)
S9 to S15 Quality of information on nursing care**	13–36	27.22 ± 4.65	3 (10)	27 (90)	0 (0)	12–36	26.22 ± 5.65	3(10)	27 (90)	0 (0)	11–37	26.22 ± 5.65	3(10)	27 (90)	0 (0)
S16 Overall quality***	3–5	3.56 ± 0.6	6 (20)	21 (70)	3 (10)	2–5	2.56 ± 0.5	7 (23.3)	21(70)	2 (6.6)	3–5	2.56 ± 0.5	7 (23.3)	21(70)	2 (6.6)
Total****	40–69	57.41 ± 5.9				40–70	58.41 ± 4.8				40–69	56.51 ± 6.8			

* Minimum-maximum scores for reliability of publication range from 8 to 40 points.** Minimum-maximum scores for quality of information on nursing care range from 7 to 35 points.*** Minimum-maximum scores for overall quality range from 1 to 5 points.**** Minimum-maximum scores of DISCERN tool range from 16 to 80 points

standardization and comparability across models, it may have constrained the depth, clinical richness, and specificity of the generated care plans. Future studies should investigate whether advanced prompt engineering strategies, such as few-shot prompting or structured templates, can improve the clinical applicability and quality of AI-generated nursing care plans.

Reliability represents a critical dimension of quality in nursing care planning, as clinical decisions are directly influenced by written content. In this study, only 25% of AI-generated references were verifiable, raising serious concerns regarding hallucinated or fabricated citations. Although AI systems such as ChatGPT have demonstrated adequate performance on standardized medical examinations, previous research has consistently reported the presence of inaccurate or fabricated references in AI-generated outputs [19, 43, 44]. This limitation was reflected in the DISCERN scores observed in the present study, which indicated only moderate reliability despite the inclusion of seemingly credible references.

From an academic integrity perspective, an important methodological finding was that traditional plagiarism detection tools (iThenticate and Turnitin) classified all AI-generated texts as original, whereas Turnitin's AI detection feature successfully identified all texts as AI-produced. This discrepancy highlights the limitations of conventional plagiarism detection systems in identifying fabricated yet original AI-generated content and underscores the growing importance of AI-specific detection tools in academic and clinical environments.

Although artificial intelligence holds significant potential to support healthcare education, clinical decision-making, and documentation, the findings of this study suggest that current AI systems should not be considered independent authorities for nursing care planning. Instead, AI-generated outputs should be used cautiously and always under professional supervision. Human judgment, clinical expertise, and ethical responsibility remain indispensable, particularly in areas requiring individualized care, emotional sensitivity, and contextual decision-making [45].

Overall, while artificial intelligence offers promising opportunities in healthcare, concerns related to readability, information reliability, and ethical use persist [46]. The present study contributes to the literature by providing a focused evaluation of AI-generated nursing care plans and emphasizes the need for transparent, verifiable, and human-centered integration of AI technologies into nursing practice.

Strength of the study

A major strength of this study is its comprehensive evaluation of AI-generated nursing care plans across multiple dimensions, including readability, quality, and reliability.

Table 6 Correlation analysis between descriptive criteria and DISCERN

Descriptive Criteria		ChatGPT Reliability of Publication	ChatGPT Quality Information on Nursing Care	ChatGPT Overall Quality	ChatGPT DISCERN, Total	Gemini Reliability of Publication	Gemini Quality Information on Nursing Care	Gemini Overall Quality	Gemini DISCERN, Total	DeepSeek Reliability of Publication	DeepSeek Quality Information on Nursing Care	DeepSeek Overall Quality	DeepSeek DISCERN, Total
Number of references	*r*	0.271	−0.013	−0.129	0.124	0.282	−0.033	−0.128	0.154	0.272	−0.043	−0.149	0.136
	P	0.104	0.829	0.453	0.440	0.123	0.838	0.452	0.420	0.123	0.838	0.463	0.413
Number of article references	*r*	0.015	−0.127	−0.073	−0.040	0.015	−0.132	−0.072	−0.060	0.006	−0.112	−0.083	−0.042
	P	0.986	0.677	0.751	0.891	0.925	0.687	0.754	0.872	0.975	0.667	0.761	0.896
Number of book references	*r*	0.170	0.023	−0.125	0.041	0.120	0.039	−0.135	0.061	0.170	0.019	−0.145	0.056
	P	0.348	0.923	0.568	0.783	0.368	0.922	0.578	0.783	0.388	0.930	0.578	0.792
Number of webreferences	*r*	0.229	0.121	0.129	0.170	0.229	0.124	0.139	0.170	0.229	0.140	0.179	0.170
	P	0.296	0.452	0.419	0.333	0.166	0.456	0.519	0.343	0.186	0.429	0.429	0.343
Word count	*r*	0.112	−0.071	0.395	0.011	0.132	−0.083	0.495	0.031	0.132	−0.063	0.385	0.031
	P	0.751	0.757	0.182	0.926	0.751	0.756	0.282	0.956	0.751	0.726	0.172	0.956
Paragraph count	*r*	0.014	0.121	−0.061	0.072	0.014	0.124	−0.061	0.082	0.034	0.153	−0.071	0.072
	P	0.887	0.522	0.758	0.561	0.826	0.522	0.858	0.561	0.876	0.537	0.768	0.581
Similarity rate	*r*	0.233	−0.131	0.096	0.038	0.242	−0.136	0.066	0.068	0.242	−0.145	0.097	0.078
	P	0.212	0.402	0.565	0.625	0.221	0.407	0.575	0.625	0.221	0.425	0.575	0.676
Number of accessible references	*r*	0.423	0.112	0.221	0.19	0.438	0.112	0.231	0.49	0.438	0.120	0.321	0.37
	P	**0.007***	0.471	0.161	**0.017***	**0.007***	0.478	0.141	**0.029***	**0.007***	0.429	0.172	**0.017***
Number of accessible article references	*r*	0.342	0.073	0.282	0.229	0.342	0.073	0.262	0.279	0.342	0.062	0.272	0.289
	P	**0.037***	0.648	0.087	0.021	**0.014***	0.644	0.077	0.081	**0.024***	0.648	0.077	0.071
Number of accessible book references	*r*	0.183	0.147	0.029	0.269	0.185	0.143	0.039	0.249	0.185	0.151	0.039	0.249
	P	0.319	0.416	0.838	0.165	0.329	0.418	0.848	0.165	0.308	0.429	0.848	0.155
Number of accessible web references	*r*	0.278	0.151	0.126	0.228	0.278	0.154	0.136	0.268	0.278	0.175	0.116	0.268
	P	0.062	0.362	0.479	0.073	0.074	0.362	0.489	0.073	0.054	0.383	0.489	0.073

**P <.05;* r: *Spearman*

Unlike studies focusing on a single AI platform, this research compared three widely used chatbots. Additionally, readability assessments were standardized using two independent and publicly accessible calculators, and inter-rater reliability for DISCERN evaluations was rigorously examined, enhancing the methodological robustness of the findings.

Limitations of the study

Several limitations should be acknowledged. First, the analysis was limited to three AI chatbots, which may restrict generalizability to other emerging AI systems. Second, all texts were generated on a single date, and AI outputs may change as models continue to evolve. Third, although the sample size of 90 texts allowed for systematic comparisons, it represents only a small proportion of the potential variability in AI-generated outputs.

An additional limitation relates to the prompt design. The use of a zero-shot, generic prompt without role specification or structural constraints may have limited the depth and clinical specificity of the generated nursing care plans. While this approach ensured comparability across models, future research should explore how structured and role-based prompts influence the quality and usability of AI-generated nursing documentation.

Conclusion

This study evaluated nursing care plan texts generated by ChatGPT, Gemini, and DeepSeek with respect to readability, quality, and reliability. The findings demonstrated that all AI-generated texts exceeded the recommended sixth-grade reading level and were generally characterized by moderate quality and reliability. Although the inclusion of accessible references was associated with improved reliability, the high prevalence of unverifiable citations and elevated literacy demands may limit the practical usability of these texts in clinical settings.

Given the critical importance of clarity and accuracy in healthcare communication, AI-generated nursing care plans should not be used as standalone resources. When applied under professional supervision, however, AI tools may serve as supportive educational aids for nursing students and productivity-enhancing resources for healthcare professionals. To improve their clinical relevance, future AI systems should incorporate up-to-date, evidence-based databases and transparent reference validation mechanisms. Nevertheless, professional judgment and human oversight remain essential to ensure the safe and appropriate integration of AI-generated content into nursing practice.

Abbreviations

ARI Automated Readability Index
CLI Coleman Liau Readability Index
FKGL Flesch-Kincaid Grade Level
FRES Flesch Reading Ease Score
GFOG Gunning Fog Index
GQS Global Quality Score
LW Linsear Write
SMOG Simple Measure of Gobbledygook

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12912-026-04295-7.

Supplementary Material 1

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Author contributions

Gokcen Gokalp M. and Cinar Yucel S. designed the study. Data were collected and analyzed by Gokcen Gokalp M. The manuscript was drafted by Gokcen Gokalp M. and reviewed by Cinar Yucel S. Both authors read and approved the final version of the manuscript.

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Data availability

The datasets generated and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

Researchers utilized AI platforms provided as a "free research preview," which did not require formal ethical approval or institutional authorization. Because the data are transparent and verifiable, the study clearly details the methods of AI use, including the selection, randomization, and organization of nursing diagnoses. Therefore, formal ethical approval was not required.

Consent for publication

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Competing interests

The authors declare no competing interests.

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